

Positive-payoff quadrature for Black–Scholes: a frozen binary64 prototype

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Result and scope

We replaced the cancellation-prone branches of our experimental v0 evaluator with a separate binary64 positive-integral implementation, candidate v2. On **288 new confirmation inputs**, all predeclared practical price/log gates passed, with no evaluation exception, wrong OTM side, negative/nonfinite price, or unexpected zero. Among **202** prices whose converged reference rounds to a nonzero binary64 value, v2 was closer than v0 in 192 cases, farther in 3, and equal in 7. Against the unchanged historical Jäckel source, it was closer in 146, farther in 30, and equal in 26.

This is an improvement over our previous numerical prototype on these tests. It is **not** a new BSM model, a proof of universal superiority, a novelty claim, or a demonstrated improvement in market prediction. The corpus deliberately emphasizes near-ATM cancellation and range limits; these counts are not market failure probabilities. A small timing check found the Python v2 detailed API about **15.4 times slower** than the Python v0 detailed API on the measured development subset. Their achieved accuracy differs, and C++ speed was not timed.

The preceding audit and its negative result remain in TECHNICAL_NOTE_V1.md.

1. Shared mathematical representation

For the same exact-binary64 input contract as v1, put $m = \log(S/K) + (r - q)T$, $v = \sigma\sqrt{T}$, and $z = v/2 - |m|/v$. The OTM leg is call for $m \leq 0$, put otherwise; its discounted cash leg is $A = Se^{-qT}$ or $A = Ke^{-rT}$, respectively. For positive total volatility,

$$V_{OTM} = \frac{A}{\sqrt{2\pi}} \int_0^\infty e^{-(u-z)^2/2} (1 - e^{-vu}) du.$$

The integrand is nonnegative. This representation removes the subtraction of two nearly equal CDF/Mills terms. It was already used as a cross-formula check in v1; here it is implemented as an ordinary binary64 candidate. No multiprecision arithmetic or oracle fallback occurs inside v2.

For $z \leq 0$, set $L = \max(1, -z)$, $b = v/L$, and $u = w/L$. We integrate

$$I = \int_0^{64} e^{(z/L)w - w^2/(2L^2) - \frac{\expm1(-bw)}{b}} dw$$

and reconstruct $V = A\phi(z)(v/L^2)I$. When rounded $b = 0$, the integrand factor takes its limit w ; the external v is retained. The omitted integral is bounded in exact arithmetic by $65e^{-64}$ for $z \leq -1$, or $e^{-64^2/2}$ for $-1 < z \leq 0$.

For $z > 0$, we integrate the completed-square Gaussian on $[\max(0, z - 12), z + 12]$, dividing the payoff factor by v and restoring v outside. The integrand is bounded by $ue^{-(u-z)^2/2}$, giving an omitted-integral bound $(1 + cz/12)e^{-72}$, where $c = 1$ if $z \leq 12$ and $c = 2$ otherwise. Bounds are evaluated in log form and checked relative to the estimated integral. Floating-point evaluation of these bounds is not directed interval arithmetic.

Both branches use adaptive Gauss–Legendre 16/32 panels. We retain positive panel sums, allocate an absolute budget across panels, and include a binary64 roundoff floor. Depth and evaluation budgets are finite; exhaustion raises an explicit exception. The node-rule difference estimates integration error; it does not bound all input, elementary-function, or rounding errors.

2. Input preparation and reconstruction

Near ATM, `v2` computes $\log_{10}((S-K)/K)$, retaining the small input difference. For separated values it uses `frexp` mantissas/exponents, avoiding overflow in S/K . Carry uses $(r-q)*T$ and `fsum` when adding log moneyness. These operations still round; we make no general sign guarantee at arbitrary near-exact carry cancellation.

Ordinary and log prices use the same computed integral and the same side. The ordinary result retains the original monetary mantissa and accumulates power-of-two exponents before final `ldexp`; an intermediate normalized price is never required to fit into binary64. The log result is a second view of these factors, not a separately selected pricing formula. We do not require bitwise equality to `exp(rounded_complete_log_price)`, which can itself lose accuracy from rounding a large monetary logarithm.

Intrinsic value uses `expm1` or the exact near-input cash difference where applicable; the ITM leg is reconstructed by positive addition. The source does not import `mpmath`, call native Jäckel, or read reference data.

The API explicitly rejects a numerically unsupported range: positive σ, T with $\sigma*\sqrt{T}$ rounded to zero. For example, with $S=K=1e200$, $\sigma=1e-300$, $T=1e-100$, the true ATM time value is approximately $3.99e-151$ even though ordinary binary64 total volatility rounds to zero. Returning a deterministic payoff would be wrong. True zero volatility or zero maturity remains supported. Nonfinite intermediate log arithmetic and insufficient integration budgets also fail explicitly.

3. Freeze and evidence

`EXPERIMENT_V2.md` defines the construction and gates. All 316 old `v1` cases were development data. A first development-only quadrature attempt exhausted a purely local relative tolerance in 105 cases; that attempt is preserved in `evidence/archive-v2-draft-01`. The global absolute budget fixed this integration-control problem before confirmation.

The final candidate, its unit-test module, protocol and oracle were hashed before any `v2` confirmation output was computed. The candidate SHA-256 is

536509e79dc52cd22af5b6dadbc4aeac9c46f573df30e522d91e6ba5a3572b78. The corpus SHA-256 is 019016a0e6a8256c4290033780d43c1226ae1ffb7bcd150b75ab23daf57d978. The freeze receipt is `evidence/freeze-v2.json`. No candidate changes were made after observing confirmation results.

The freeze receipt did not include the complete executable pipeline: the final runner hash is recorded in its output, while some imported helper modules were not in the pre-view receipt. Thus we claim a locally frozen candidate and corpus with an auditable final runner, not a pre-view cryptographic freeze of every executed byte or independent timestamped preregistration.

The primary reference is the unchanged exact-binary `mpmath erfc` formula, evaluated independently at 100/180 digits with the registered 260-digit retry available. All 288 cases passed without that retry. Ten predetermined points also passed a 100-digit positive-integral check, with maximum absolute log disagreement about $1.30e-91$. This supplementary check shares the analytic integral with `v2`, while its arithmetic and integration routine differ; it must not be portrayed as an entirely independent mathematical formula for `v2`. The primary `erfc` reference remains the cross-formula comparison.

All **22 unit/property tests** passed. They include polynomial moments for the quadrature rules, an exact ATM `erf` identity at several monetary scales, parity, symmetry, monotonicity, range reconstruction, `v1`

regressions and the explicit total-volatility-underflow rejection. The frozen candidate also reproduces all 316 measured development prices/logs and passes their gates.

4. New confirmation results

There are 201 normal, 1 nonzero subnormal, and 86 zero-rounding reference prices. The table excludes the latter 86 trivial zero comparisons.

New input family	Nonzero cases	v2 closer than Jäckel	Jäckel closer	Equal error
Near-ATM monetary scales	96	88	2	6
Normalized mixture	82	41	21	20
Nonzero carry	13	9	4	0
Tail and monetary range	11	8	3	0
Total	202	146	30	26

Counts compare absolute monetary error against the converged reference. Nonzero-carry rows remain end-to-end BSM comparisons; rounded-forward mapping and subsequent evaluation errors are recorded separately. For zero carry alone, the comparison is 137/26/26 across 189 nonzero cases. This is still not an isolated normalized-kernel comparison.

Diagnostic	Frozen v0	Candidate v2	Historical Jäckel
Unexpected zero among 202 nonzero cases	1	0	1
Matches rounded reference among those cases	8	73	27
Largest relative error, normal prices	1	1.19e-13	1
Largest relative error, all nonzero prices	35.22	2.45e-13	1

Rounded-reference matches use an empirical converged reference, not a formal correct-rounding certificate. All v2 outputs passed the predeclared monetary budget $\max(1e-10 \cdot \text{reference}, \text{one ULP at rounded reference})$. All log outputs passed $\max(2e-11, 16 \text{ ULP at rounded reference log})$; this is an absolute budget with a representational allowance, not uniformly small relative log error.

For example, at v2-0163 the log error is only about $1.20e-16$ in absolute units, but roughly 2.26 million ULP because the reference log is near zero. Conversely, in an extremely deep zero-price tail (v2-0236), absolute log error is about 0.00237 while the log's own spacing is coarse. Passing the mixed budget must not be summarized as all log errors being below $2e-11$.

In v2-0272, the reference OTM price is about $2.550803296012454e-241$. Both v0 and historical Jäckel return zero; v2 returns approximately $2.550803296012165e-241$, relative error $1.14e-13$. In v2-0274, v2 matches the rounded subnormal reference around $4.480857576181261e-312$; v0's relative error is about 35.22 and Jäckel's about 0.00108. These are synthetic range probes, not demonstrated losses on real positions, and novelty has not been established.

The 3 cases where v0 is closer (v2-0138, v2-0155, v2-0183) and all 30 Jäckel wins are retained. Nothing was removed to improve a score.

5. Cost and practical role

A post-accuracy timing check used the 106 old fixtures at indices 0,3,6,..., one warm-up batch per implementation, five timed batches and alternating method order. Median per-call times on this macOS arm64 machine were about 5.93 microseconds for Python `stable_detailed` v0 and 91.34 microseconds for Python v2, ratio 15.4. This compares the actual detailed APIs at unequal achieved accuracy. It excludes reference calculations, compilation and file I/O; it says nothing about native C++ throughput or production latency.

The most expensive confirmation call used 480 integrand evaluations. All computation was sequential on one numerical CPU thread, with no GPU. This v2 is a useful accuracy-oriented independent implementation/fallback candidate, not yet a replacement for a high-throughput pricing library.

6. Limitations and next step

The Jäckel baseline remains the unchanged historical [vollib commit 83ae882](#), associated with **Let's Be Rational**. The latest official implementation has not been verified. This note makes no new discovery claim about that source or the established positive integral.

The reference is empirical rather than interval-certified. Gates cover this corpus and tolerate much more than one ULP for some normal prices. The near-zero log regime and tiny-total-volatility range remain explicit limitations. We have not established full binary64-domain coverage, smooth Greeks, reliable IV inversion, model calibration, hedging gains, or production suitability. Internal AI review is not independent human peer review.

The next research step is independent reproduction of this frozen package, then a separately versioned, condition-controlled fast path with the integral as a fallback. Both v1 and v2 corpora are now consumed; a new accuracy claim for another candidate requires a new confirmation set. Keep the slow v2 as an inspectable reference rather than silently changing it for a better score.

Full rows: `evidence/benchmark-v2.json`. Timing: `evidence/python-timing-v2.json`. Source: `lab/black_scholes_v2.py`. Reproduction commands are in `README.md`.

Authors and research transparency

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Khamit initiated and directed the research programme. Daniyal is a coauthor and the designated lead for its continuing software development. That prospective responsibility does not assert that he independently implemented or validated the frozen v2 code. The recorded experiment was prepared and executed with OpenAI Codex assistance, including mathematical analysis, implementation, test generation, result checking, drafting and editorial review. AI-role reviews are not human peer review. The work is an interim, non-peer-reviewed computational study; independent reproduction and human technical review are invited. No academic or financial institution endorsement is claimed.

References

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https://github.com/vollib/lets_be_rational/tree/83ae882df8e19323798c7ebfb8898f94d2d92ade
4. `mpmath` 1.3.0, arbitrary-precision reference arithmetic. <https://mpmath.org/>